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Batch: PG-DAC February 2026
Subject: JAVA
Assignment: 2
Form Number: 251101107

// 1. Write a program to check if a given number is positive, negative, or zero.

Solution:

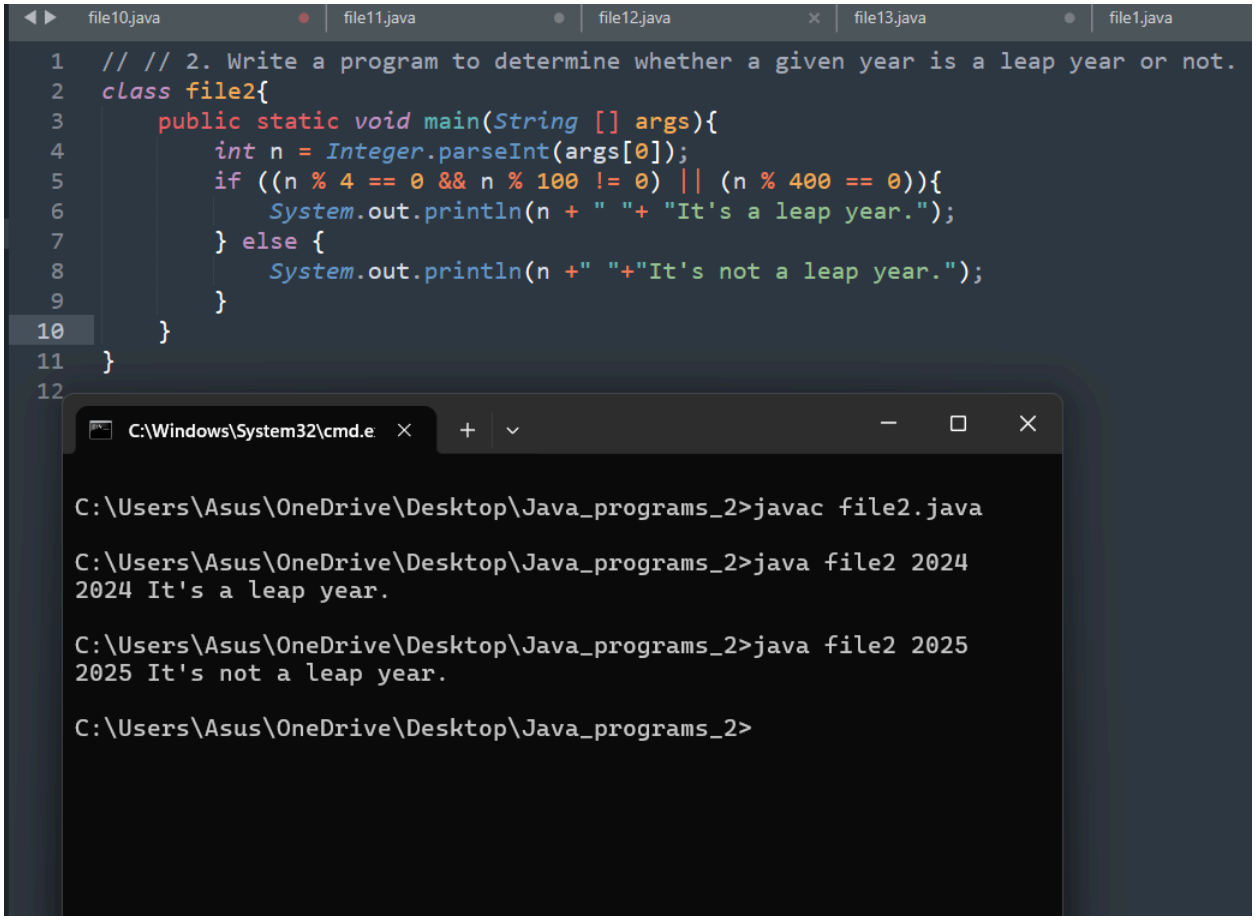
```
class file1{
	public static void main(String [] args){
		int n = Integer.parseInt(args[0]);

		if (n > 0){
			System.out.println(n + " Is a Positive Number");
		}
		else if(n < 0){
			System.out.println(n + " Is a Negative Number");
		}
		else{
			System.out.println(n + " Is Zero");
		}
	}
}
```

```java
// 1. Write a program to check if a given number is positive, negative, or z

class file1{
    public static void main(String [] args){
        int n = Integer.parseInt(args[0]);

        if (n > 0){
            System.out.println(n + " Is a Positive Number");
        }
        else if(n < 0){
            System.out.println(n + " Is a Negative Number");
        }
        else{
            System.out.println(n + " Is Zero");
        }
    }
}
```

```
C:\Windows\System32\cmd.e

C:\Users\Asus\OneDrive\Desktop\Java_programs_2>javac file1.java

C:\Users\Asus\OneDrive\Desktop\Java_programs_2>java file1 0
0 Is Zero

C:\Users\Asus\OneDrive\Desktop\Java_programs_2>java file1 1
1 Is a Positive Number

C:\Users\Asus\OneDrive\Desktop\Java_programs_2>java file1 -1
-1 Is a Negative Number
```

// // 2. Write a program to determine whether a given year is a leap year or not.
Solution:

```java
class file2{
    public static void main(String [] args){
        int n = Integer.parseInt(args[0]);
        if ((n % 4 == 0 && n % 100 != 0) || (n % 400 == 0)){
            System.out.println(n + " "+ "It's a leap year.");
        } else {
            System.out.println(n +" "+"It's not a leap year.");
        }
    }
}
```



// 3. Write a program to check if a given character is a vowel or a consonant.

Solution:

```
class file3{
        public static void main(String []args){
                char n = args[0].charAt(0);
                if (n == 'a'|| n ==  'e' || n == 'i' || n == 'o' || n =='u'){
                        System.out.println(n +" " +"It's a vowel.");
                }else{
                        System.out.println(n + " "+"It's a consonant");
                }
        }
}
```

```java
// 3. Write a program to check if a given character is a vowel or a consonant.
class file3{
    public static void main(String []args){
        char n = args[0].charAt(0);
        if (n == 'a'|| n ==  'e' || n == 'i' || n == 'o' || n =='u'){
            System.out.println(n +" " +"It's a vowel.");
        }else{
            System.out.println(n + " "+"It's a consonant");
        }
    }
}
```

```
C:\Windows\System32\cmd.e

C:\Users\Asus\OneDrive\Desktop\Java_programs_2>javac file3.java

C:\Users\Asus\OneDrive\Desktop\Java_programs_2>java file3 a
a It's a vowel.

C:\Users\Asus\OneDrive\Desktop\Java_programs_2>java file3 s
s It's a consonant

C:\Users\Asus\OneDrive\Desktop\Java_programs_2>
```

// 4. Write a program to find the largest of three numbers entered by the user.
Solution:

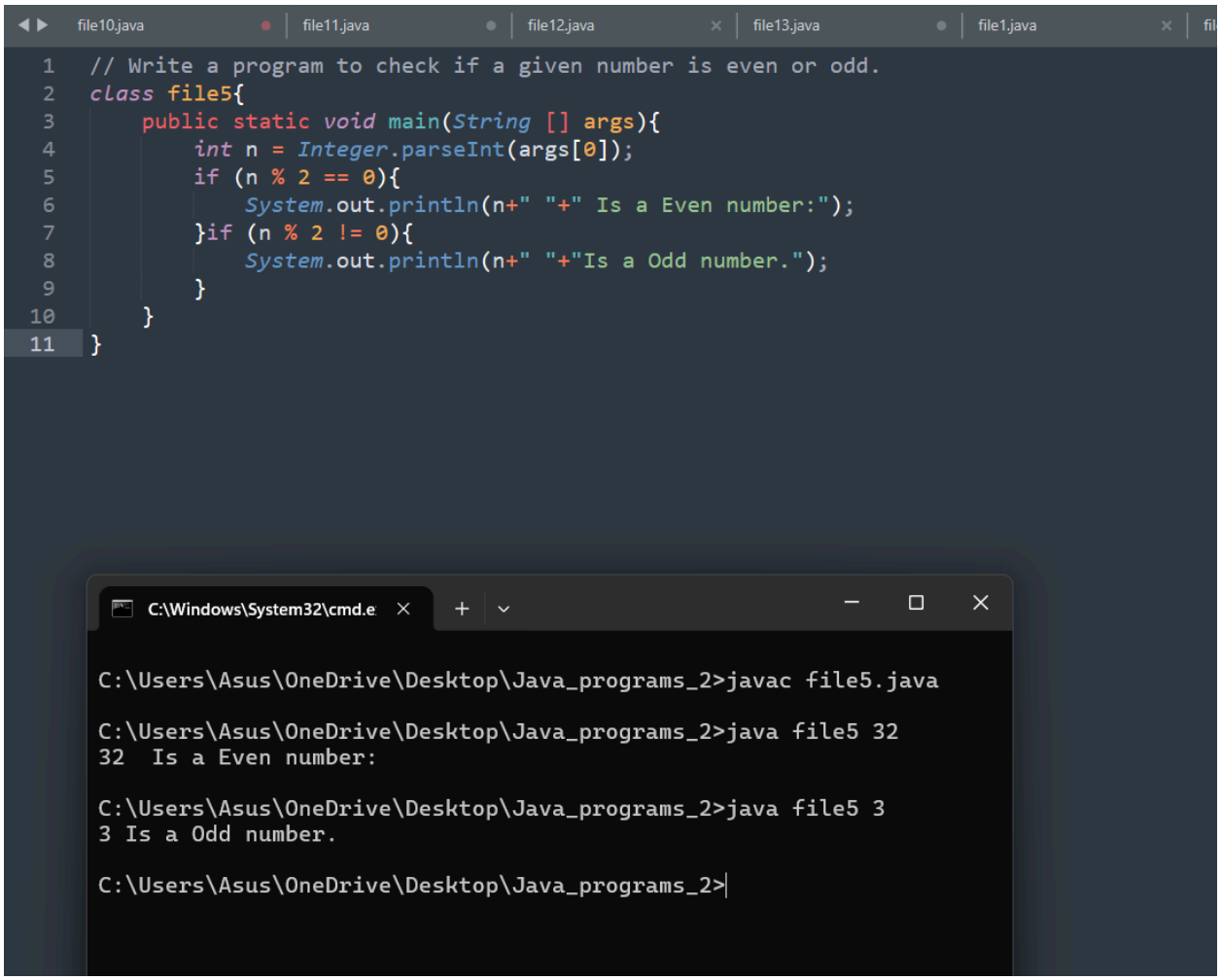
```java
class file4{
        public static void main(String []args){
                int n1 = Integer.parseInt(args[0]);
                int n2 = Integer.parseInt(args[1]);
                int n3 = Integer.parseInt(args[2]);
                int temp = 0;

                if (n1 > n2 && n1 >n3){
                        System.out.println(n1 +" "+"It's a greatest of all three.");
         }else if (n2>n1 && n2>n3){
                System.out.println(n2+ " "+"It's a greatest of all three.");
                }else{
                System.out.println(n3+" "+"It's a greatest of all three.");
         }
        }
}
```

```java
// 4. Write a program to find the largest of three numbers entered by the user.

class file4{
    public static void main(String []args){
        int n1 = Integer.parseInt(args[0]);
        int n2 = Integer.parseInt(args[1]);
        int n3 = Integer.parseInt(args[2]);
        int temp = 0;

        if (n1 > n2 && n1 >n3){
            System.out.println(n1 +" "+"It's a greatest of all three.");
      }else if (n2>n1 && n2>n3){
        System.out.println(n2+ " "+"It's a greatest of all three.");
        }else{
        System.out.println(n3+" "+"It's a greatest of all three.");
      }
    }
}
```

```
C:\Windows\System32\cmd.e

C:\Users\Asus\OneDrive\Desktop\Java_programs_2>javac file4.java

C:\Users\Asus\OneDrive\Desktop\Java_programs_2>java file4 21 23 0
23 It's a greatest of all three.

C:\Users\Asus\OneDrive\Desktop\Java_programs_2>java file4 43 2 9
43 It's a greatest of all three.

C:\Users\Asus\OneDrive\Desktop\Java_programs_2>
```

//5. Write a program to check if a given number is even or odd.
Solution:

```java
class file5{
        public static void main(String [] args){
                int n = Integer.parseInt(args[0]);
                if (n % 2 == 0){
                        System.out.println(n+" "+" Is a Even number:");
                }if (n % 2 != 0){
                        System.out.println(n+" "+"Is a Odd number.");
                }
        }
}
```



//6. Problem: Days of the Week
// Task: Write a program that takes an integer input (1-7) from the user and prints the corresponding day of the week.
// Use a switch statement to handle the conversion

Solution:

```
class file6{
        public static void main(String []args){
                int day = Integer.parseInt(args[0]);
                switch(day){
                case 1:
                        System.out.println("It's Monday.");
                        break;
                case 2:
                        System.out.println("It's Tuesday.");
```

```
                break;
            case 3:
                System.out.println("It's Wednesday.");
                break;
            case 4:
                System.out.println("It's Thursday.");
                break;
            case 5:
                System.out.println("It's friday.");
                break;
            case 6:
                System.out.println("It's Saturday.");
                break;
            case 7:
                System.out.println("It's Sunday.");
                break;
            default:
                System.out.println("It's Invalid day.");
            }

        }
}
```

file10.java | file11.java | file12.java | file13.java | file1.java | file2.java | file6.java

```
//6.  Problem : Days of the Week
// Task: Write a program that takes an integer input (1-7) from the user and prints the corresponding day of the week.
// Use a switch statement to handle the conversion


class file6{
    public static void main(String []args){
        int day = Integer.parseInt(args[0]);
        switch(day){
        case 1:
            System.out.println("It's Monday.");
            break;
        case 2:
            System.out.println("It's Tuesday.");
            break;
        case 3:
            System.out.println("It's Wednesday.");
            break;
        case 4:
            System.out.println("It's Thursday.");
            break;
        case 5:
            System.out.println("It's friday.");
            break;
        case 6:
            System.out.println("It's Saturday.");
            break;
        case 7:
            System.out.println("It's Sunday.");
            break;
        default:
            System.out.println("It's Invalid day.");
        }

    }
}
```

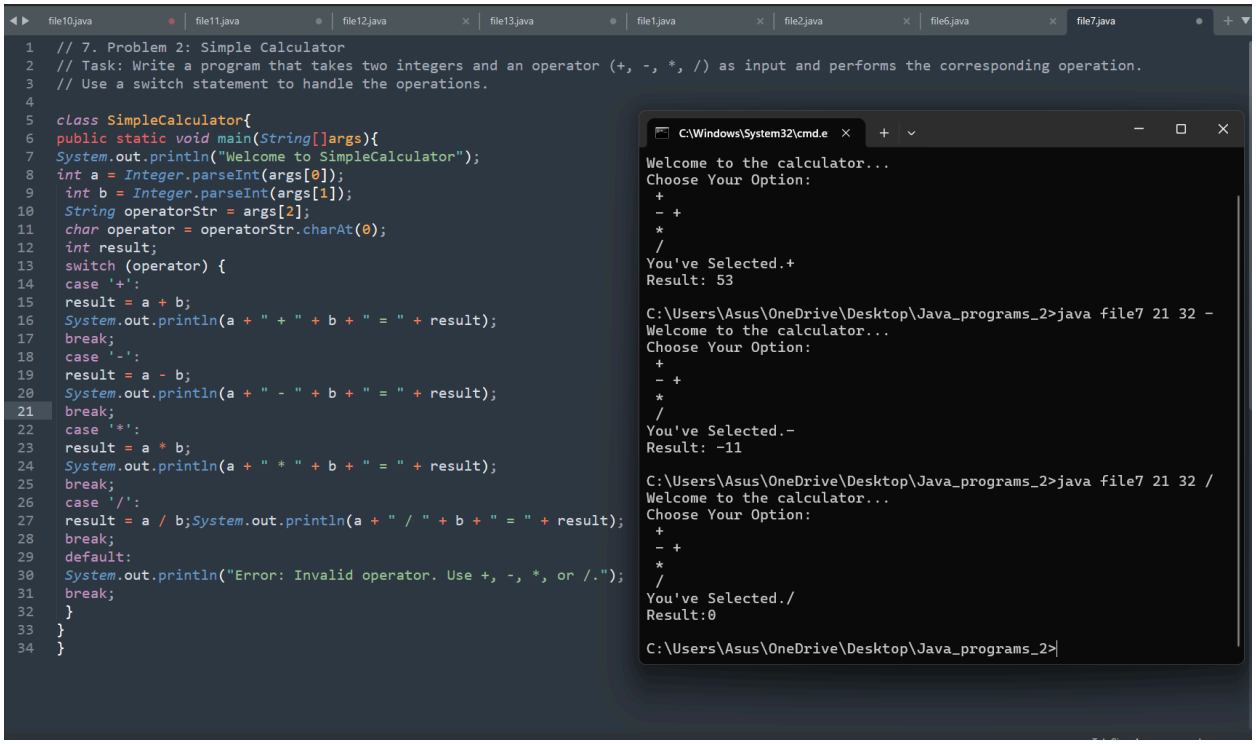
C:\Windows\System32\cmd.e

```
C:\Users\Asus\OneDrive\Desktop\Java_programs_2>javac file6.java

C:\Users\Asus\OneDrive\Desktop\Java_programs_2>java file6 1
It's Monday.

C:\Users\Asus\OneDrive\Desktop\Java_programs_2>java file6 5
It's friday.

C:\Users\Asus\OneDrive\Desktop\Java_programs_2>java file6 6
It's Saturday.

C:\Users\Asus\OneDrive\Desktop\Java_programs_2>java file6 0
It's Invalid day.

C:\Users\Asus\OneDrive\Desktop\Java_programs_2>
```

// 7. Problem 2: Simple Calculator
// Task: Write a program that takes two integers and an operator (+, -, *, /) as input and performs the corresponding operation.
// Use a switch statement to handle the operations.

Solution:

```
class SimpleCalculator{
public static void main(String[]args){
System.out.println("Welcome to SimpleCalculator");
int a = Integer.parseInt(args[0]);
 int b = Integer.parseInt(args[1]);
 String operatorStr = args[2];
 char operator = operatorStr.charAt(0);
 int result;
 switch (operator) {
 case '+':
 result = a + b;
 System.out.println(a + " + " + b + " = " + result);
 break;
 case '-':
 result = a - b;
 System.out.println(a + " - " + b + " = " + result);
 break;
 case '*':
 result = a * b;
 System.out.println(a + " * " + b + " = " + result);
 break;
 case '/':
 result = a / b;System.out.println(a + " / " + b + " = " + result);
 break;
 default:
 System.out.println("Error: Invalid operator. Use +, -, *, or /.");
 break;
 }
}
}
```

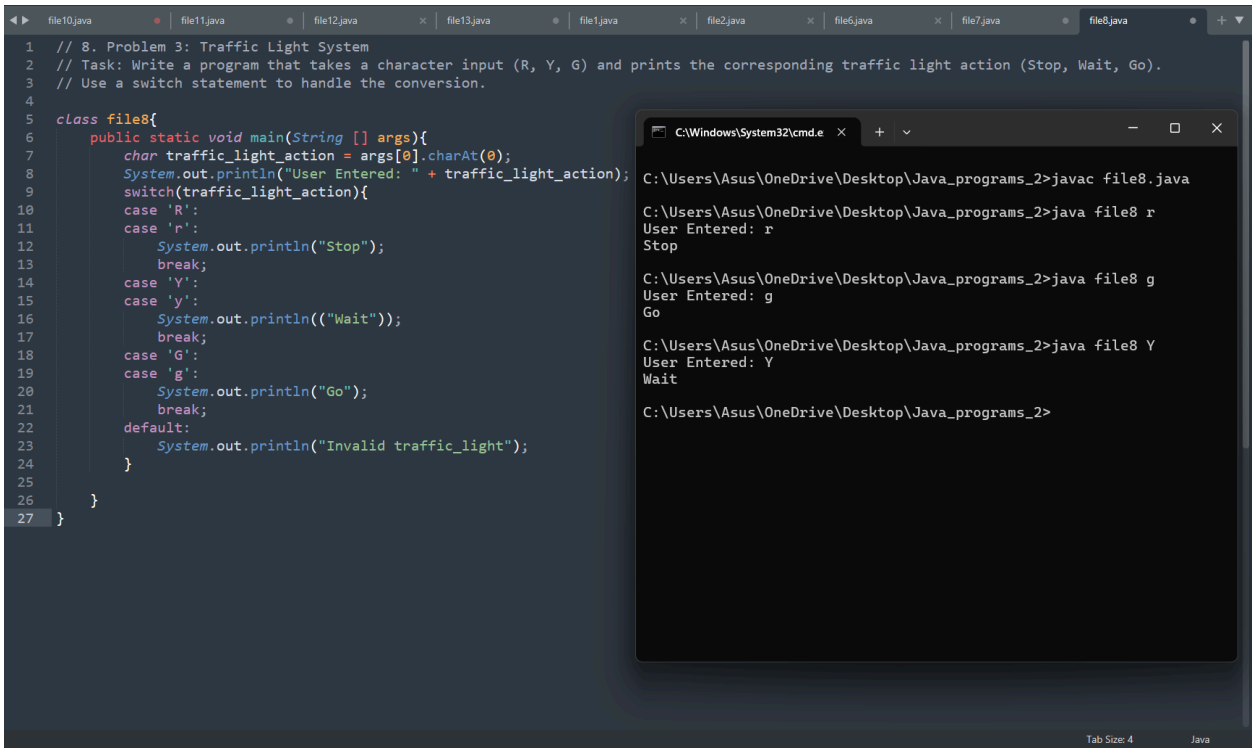


// 8. Problem 3: Traffic Light System
// Task: Write a program that takes a character input (R, Y, G) and prints the corresponding traffic light action (Stop, Wait, Go).
// Use a switch statement to handle the conversion.

Solution:

```
class file8{
        public static void main(String [] args){
                char traffic_light_action = args[0].charAt(0);
                System.out.println("User Entered: " + traffic_light_action);
                switch(traffic_light_action){
                case 'R':
                case 'r':
                        System.out.println("Stop");
                        break;
                case 'Y':
                case 'y':
                        System.out.println(("Wait"));
                        break;
                case 'G':
                case 'g':
```

```
                System.out.println("Go");
                break;
            default:
                System.out.println("Invalid traffic_light");
            }

        }
}
```

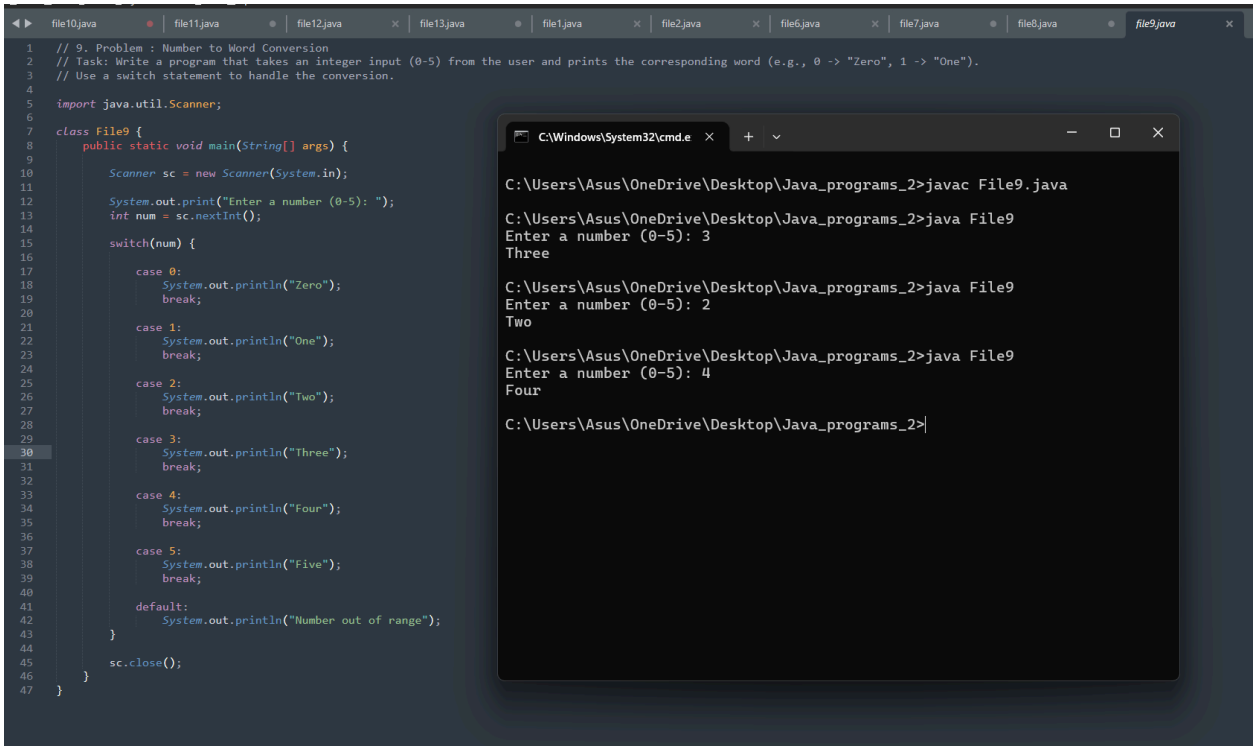


// 9. Problem: Number to Word Conversion
// Task: Write a program that takes an integer input (0-5) from the user and prints the corresponding word (e.g., 0 -> "Zero", 1 -> "One").
// Use a switch statement to handle the conversion.

Solution:

```
import java.util.Scanner;

class File9 {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter a number (0-5): ");
        int num = sc.nextInt();

        switch(num) {

            case 0:
                System.out.println("Zero");
                break;

            case 1:
                System.out.println("One");
                break;

            case 2:
                System.out.println("Two");
                break;

            case 3:
                System.out.println("Three");
                break;

            case 4:
                System.out.println("Four");
                break;

            case 5:
                System.out.println("Five");
                break;

            default:
                System.out.println("Number out of range");
        }
    }
}
```



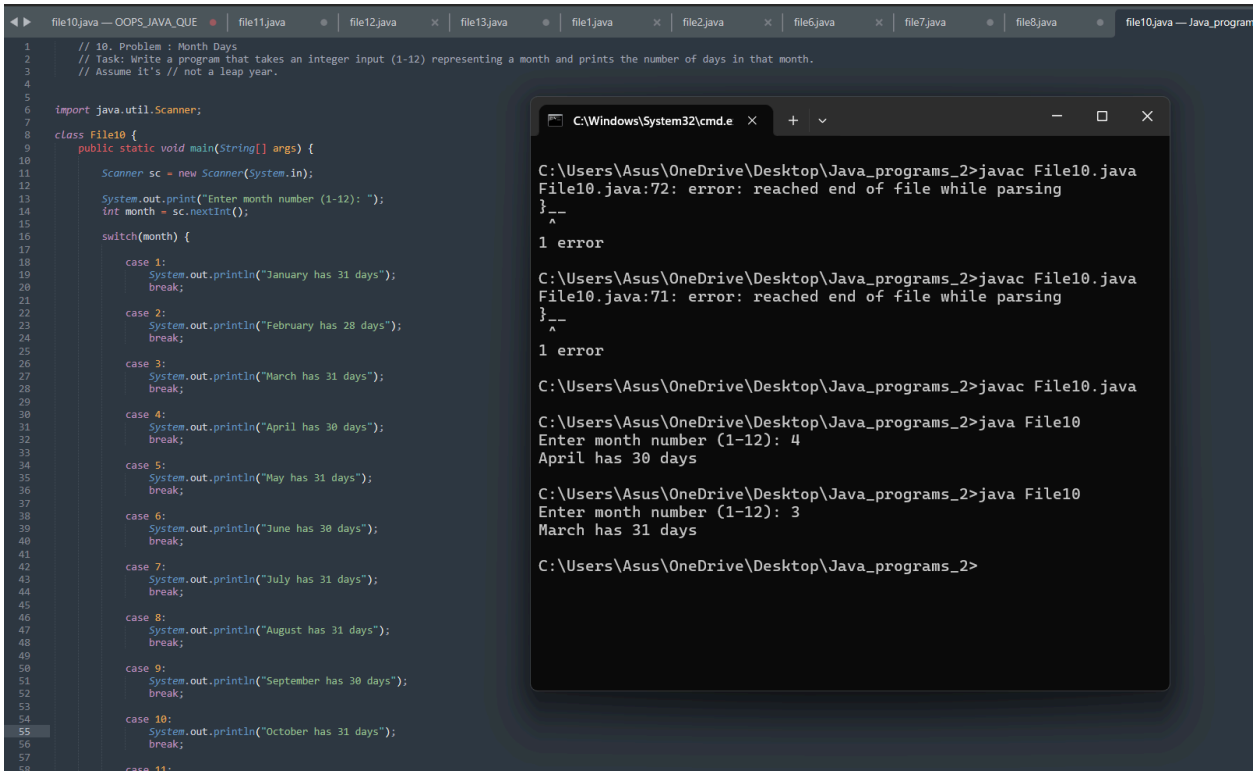
// 10. Problem: Month Days
// Task: Write a program that takes an integer input (1-12) representing a month and prints the number of days in that month.
// Assume it's // not a leap year.

Solution:

```
import java.util.Scanner;

class File10 {
   public static void main(String[] args) {

       Scanner sc = new Scanner(System.in);

       System.out.print("Enter month number (1-12): ");
       int month = sc.nextInt();

       switch(month) {

          case 1:
             System.out.println("January has 31 days");
             break;
```

```
case 2:
   System.out.println("February has 28 days");
   break;

case 3:
   System.out.println("March has 31 days");
   break;

case 4:
   System.out.println("April has 30 days");
   break;

case 5:
   System.out.println("May has 31 days");
   break;

case 6:
   System.out.println("June has 30 days");
   break;

case 7:
   System.out.println("July has 31 days");
   break;

case 8:
   System.out.println("August has 31 days");
   break;

case 9:
   System.out.println("September has 30 days");
   break;

case 10:
   System.out.println("October has 31 days");
   break;

case 11:
   System.out.println("November has 30 days");
   break;

case 12:
   System.out.println("December has 31 days");
   break;
```

```
        default:
            System.out.println("Invalid month number");
    }
  }
}__
```

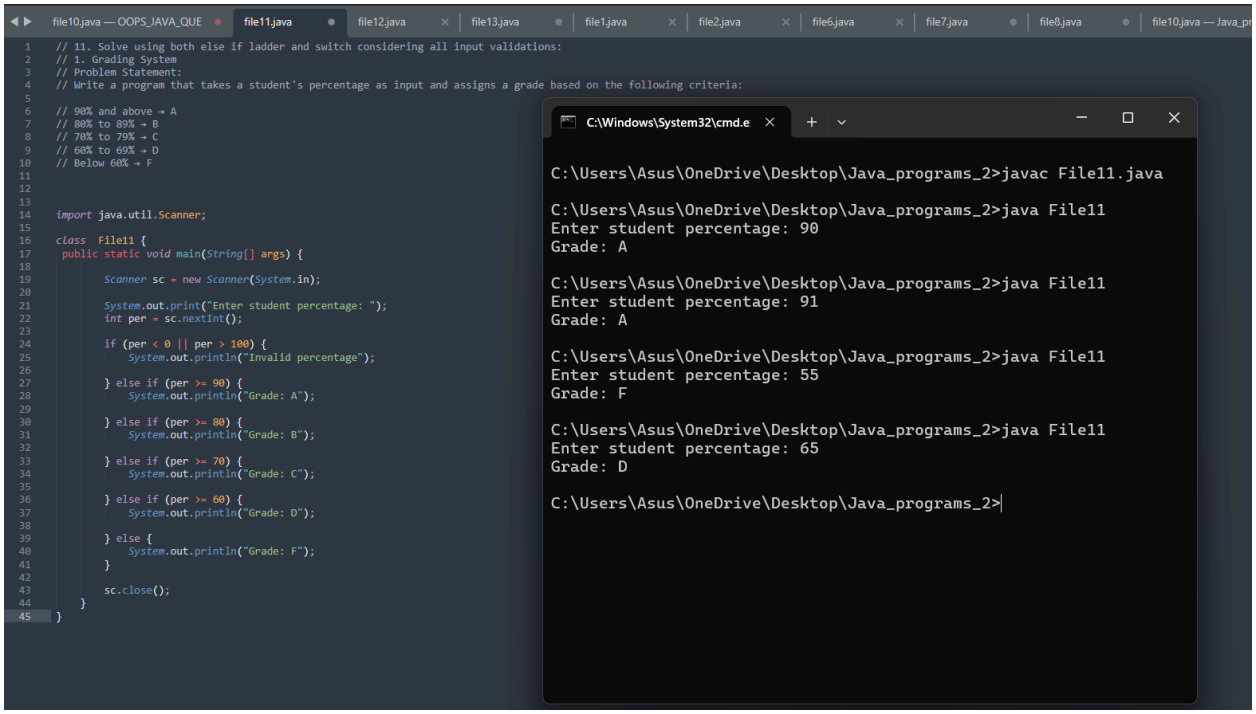


// 11. Solve using both else if ladder and switch considering all input validations:
// 1. Grading System
// Problem Statement:
// Write a program that takes a student's percentage as input and assigns a grade based on the following criteria:

// 90% and above → A
// 80% to 89% → B
// 70% to 79% → C
// 60% to 69% → D
// Below 60% → F

Solution:

```
import java.util.Scanner;

class  File11 {
 public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter student percentage: ");
        int per = sc.nextInt();

        if (per < 0 || per > 100) {
            System.out.println("Invalid percentage");

        } else if (per >= 90) {
            System.out.println("Grade: A");

        } else if (per >= 80) {
            System.out.println("Grade: B");

        } else if (per >= 70) {
            System.out.println("Grade: C");

        } else if (per >= 60) {
            System.out.println("Grade: D");

        } else {
            System.out.println("Grade: F");
        }
    }
}
```



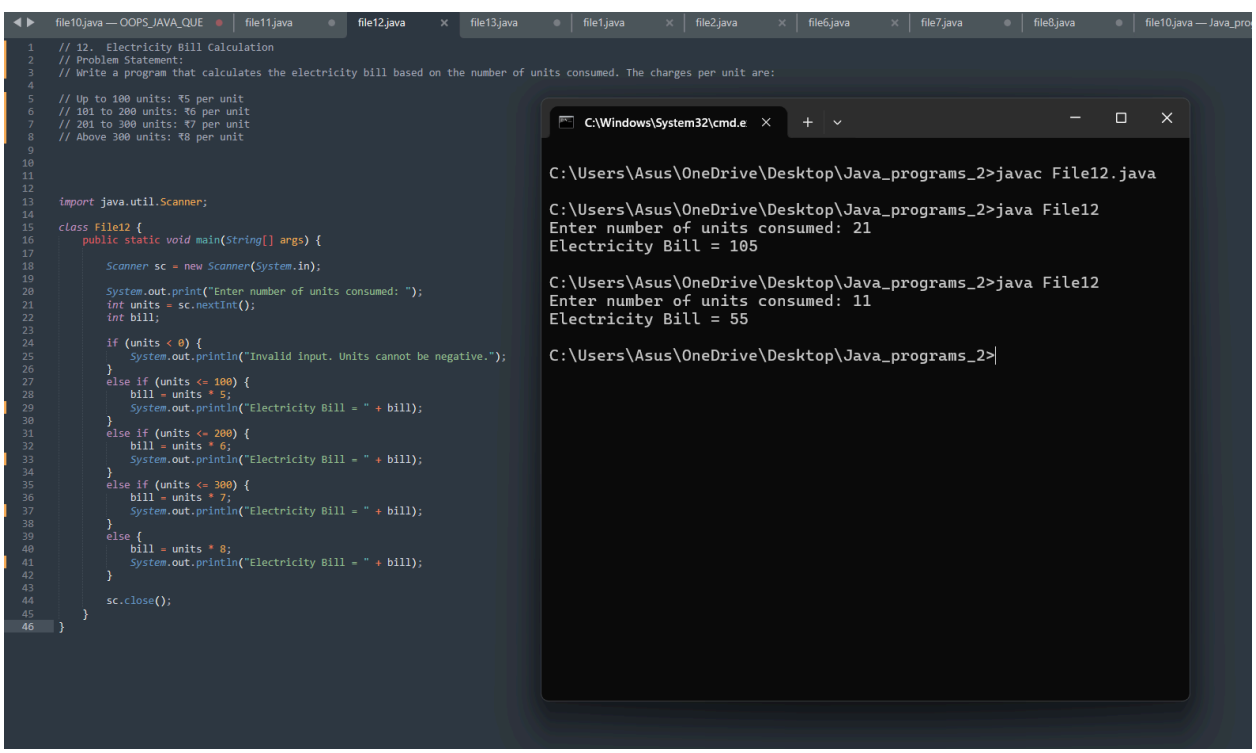
// 12. Electricity Bill Calculation
// Problem Statement:
// Write a program that calculates the electricity bill based on the number of units consumed. The charges per unit are:

// Up to 100 units: ₹5 per unit
// 101 to 200 units: ₹6 per unit
// 201 to 300 units: ₹7 per unit
// Above 300 units: ₹8 per unit

Solution:

```
import java.util.Scanner;

class File12 {
   public static void main(String[] args) {

       Scanner sc = new Scanner(System.in);

       System.out.print("Enter number of units consumed: ");
       int units = sc.nextInt();
```

```
        int bill;

        if (units < 0) {
            System.out.println("Invalid input. Units cannot be negative.");
        }
        else if (units <= 100) {
            bill = units * 5;
            System.out.println("Electricity Bill = ₹" + bill);
        }
        else if (units <= 200) {
            bill = units * 6;
            System.out.println("Electricity Bill = ₹" + bill);
        }
        else if (units <= 300) {
            bill = units * 7;
            System.out.println("Electricity Bill = ₹" + bill);
        }
        else {
            bill = units * 8;
            System.out.println("Electricity Bill = ₹" + bill);
        }
    }
}
```



// 13. Income Tax Calculation
// Problem Statement:
// Write a program that calculates the income tax payable based on the annual salary:

// Income ≤ ₹2,50,000 → No tax
// ₹2,50,001 to ₹5,00,000 → 5% tax
// ₹5,00,001 to ₹10,00,000 → 20% tax
// Above ₹10,00,000 → 30%tax

Solution:

```
import java.util.Scanner;

class File13 {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter annual income: ");
        double income = sc.nextDouble();
        double tax = 0;

        if (income < 0) {
            System.out.println("Invalid income.");
        }
        else if (income <= 250000) {
            tax = 0;
        }
        else if (income <= 500000) {
            tax = income * 0.05;
        }
        else if (income <= 1000000) {
            tax = income * 0.20;
        }
        else {
            tax = income * 0.30;
        }

        System.out.println("Tax to be paid = " + tax);

    }
}
```

file10.java — OOPS_JAVA_QUE | file11.java | file12.java | file13.java | file1.java | file2.java | file6.java | file7.java | file8.java | file10.java — Java_program

```java
// 13. Income Tax Calculation
// Problem Statement:
// Write a program that calculates the income tax payable based on the annual salary:

// Income ≤ ₹2,50,000 → No tax
// ₹2,50,001 to ₹5,00,000 → 5% tax
// ₹5,00,001 to ₹10,00,000 → 20% tax
// Above ₹10,00,000 → 30%tax


import java.util.Scanner;

class File13 {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter annual income: ");
        double income = sc.nextDouble();
        double tax = 0;

        if (income < 0) {
            System.out.println("Invalid income.");
        }
        else if (income <= 250000) {
            tax = 0;
        }
        else if (income <= 500000) {
            tax = income * 0.05;
        }
        else if (income <= 1000000) {
            tax = income * 0.20;
        }
        else {
            tax = income * 0.30;
        }

        System.out.println("Tax to be paid = " + tax);

    }
}
```

C:\Windows\System32\cmd.e

```
C:\Users\Asus\OneDrive\Desktop\Java_programs_2>javac File13.java

C:\Users\Asus\OneDrive\Desktop\Java_programs_2>java File13
Enter annual income: 200000
Tax to be paid = 0.0

C:\Users\Asus\OneDrive\Desktop\Java_programs_2>java File13
Enter annual income: 1200000
Tax to be paid = 360000.0

C:\Users\Asus\OneDrive\Desktop\Java_programs_2>java File13
Enter annual income: 700000
Tax to be paid = 140000.0

C:\Users\Asus\OneDrive\Desktop\Java_programs_2>
```